

motor. In principle, BLDC motor is controlled by means of an electronic six-step commutation system. Six-step commutation is a cost-effective means of electronic commutation as it uses simple and relatively inexpensive feedback and drive devices.

In six-step commutation, only two of the three BLDC motor windings are used at a time. Steps are equivalent to 60 electrical degrees, so six steps make a full 360-degree rotation.

Speed and torque of the motor depend on the strength of the magnetic field generated by energised windings of the motor, which depends on the current through them. Therefore adjusting the rotor voltage will change the motor speed. When using pulse-width modulation (PWM) outputs to control the six switches (of the three-phase bridge), variation of the motor voltage can be easily achieved by changing the duty cycle of PWM signal.

BLDC fan

The amount of heat generated by today's portable and feature-rich electronic products is a growing problem. Use of cooling fan remains the most common and effective way of preventing electronic



Fig. 3: BLDC motor with integrated driver circuit board

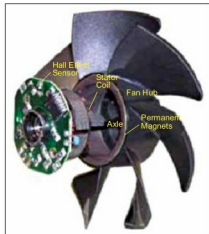


Fig. 4: Typical construction of the BLDC fan

equipment from overheating. BLDC fans are now used to fulfil the rapidly changing electronic products specifications.

The traditional DC fan has a higher inertia due to the bulky rotor coils. BLDC fan, on the other hand, has a lighter rotor due to its permanent magnets. This fan has a higher efficiency as more energy can be transferred to the load because of the lighter rotor. It uses a classic two-phase stator and rotor having surface-mounted permanent magnets. Due to the polarity reversal performed by transistors/field-effect transistors (FETs) switching in synchronisation with the rotor position, a Hall Effect sensor is used to sense the actual rotor position.

Hall-effect sensor with external or built-in driver is used in fixed-speed control of a BLDC fan. In the external method, a single output latching the three-pin Hall-effect sensor with external transistors/FETs is used. The sensor detects the polarity of the mag-

netic field generated by the permanent magnets on the rotor and provides information about the position of the rotor. This information is used to switch the external transistors/FETs. As the transistors are operated, BLDC fan will rotate continuously at a speed depending on the supplied voltage.

Built-in driver method is an enhancement over fixed-speed control version. Here the Hall sensor not only includes on-chip Hall amplifier circuitry as three-pin Hall sensor but also two complementary outputs for driving two-phase BLDC fan's coil and some extra features such as auto shutdown and restart.

Sensorless drive!

In general, a BLDC motor drive uses one or more Hall sensors giving positional information to keep synchronisation. Such implementation results in higher drive cost due to sensor wiring and implementation in the motor.

Sensorless drive is based on the detection of back electromotive force (B-EMF) induced by the movement of the permanent magnet rotor in front of stator winding, and uses the zero crossing of B-EMF to synchronise phase commutations. However, this requires use of a trapezoidal signal in order to have zero-crossing of the B-EMF (a sine wave signal drive does not provide zero-crossing signals). To detect B-EMF, a specific 120-degree six-step drive is commonly used. This drive forces zero current twice in each phase during the six-step period. This allows B-EMF zero-crossing to be precisely observed and read.

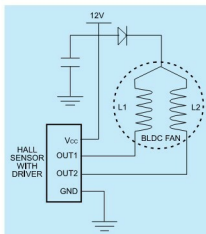


Fig. 5: Hall sensor with built-in drivers



Fig. 6: BLDC motor controller with PWM sensorless drive and on-board three-phase motor driver (Courtesy: Future Designs, Inc.)

Microcontroller solution

Microcontrollers are becoming common in motor control applications by replacing application-specific integrated circuits (ASICs). The cost of microcontrollers has come down and many manufacturers have started selling microcontroller series suitable for BLDC motor control applications.

In a microcontroller-based driver solution, many speed control algorithms can be adopted. Typical